Interference between Satellite Circuits

1. Introduction

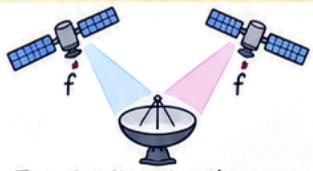
- When many satellite circuits (transponders) are used in the same frequency band, the signal from one circuit may interfere with another circuit.
- This unwanted mixing of signals is called interference between satellite circuits.
- Interference reduces the quality of communication and increases bit errors. Therefore, proper frequency planning and isolation are required.

2. What is Interference?

- Interference is the unwanted signal that enters the receiver along with the desired signal.
- It can come from:
  - another satellite circuit (same satellite),
  - a neighbouring satellite,
  - other earth stations.
- It causes reduction in carrier to interference ratio (C/I) and degrades the received signal quality.
- If interference is high, it leads to errors in voice, video or data transmission.

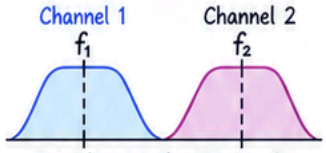
3. Types of Interference

① Co-channel Interference



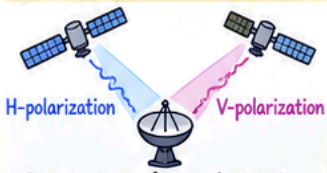
Two circuits using the same frequency.

② Adjacent channel interference



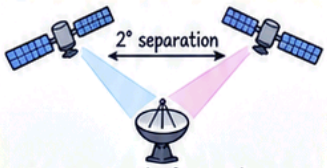
Interference due to signals in nearby frequency channels.

③ Cross-polarization interference



Due to imperfect polarization isolation.

④ Neighbouring satellite interference

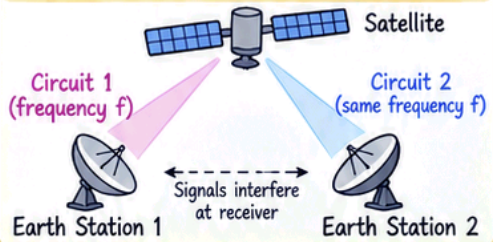


Due to signals from adjacent satellites.

4. Co-channel Interference

- When two satellite circuits use the same frequency channel, their signals overlap at the receiver.
- This causes co-channel interference.
- It is mainly due to:
  - Reuse of the same frequency band in another beam or satellite.
  - Insufficient isolation between circuits.
- It reduces C/I ratio and increases bit errors.

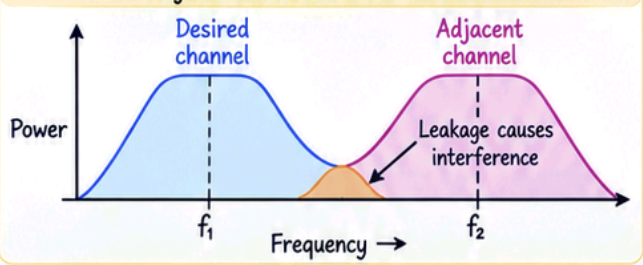
Co-channel Interference Diagram



5. Adjacent Channel Interference

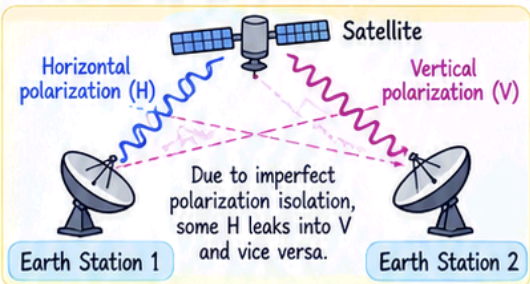
- Occurs when a strong signal in one channel leaks into the neighbouring channel.
- It is due to imperfect filters and non-ideal transmitter and receiver characteristics.
- It causes interference even though different frequencies are used.
- Proper guard band and high-quality filters are used to reduce this.

Adjacent Channel Interference



6. Cross-Polarization Interference

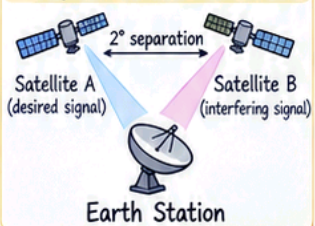
- Satellites use orthogonal polarizations (Horizontal and Vertical) to increase frequency reuse.
- In practice, the isolation between the two polarizations is not perfect.
- Due to cross-polarization discrimination (XPD) limitations, a small portion of the signal leaks into the orthogonal polarization and causes interference.



7. Interference from Neighbouring Satellites

- Satellites are placed in geostationary orbit at fixed longitudes separated by about 2°.
- If the antenna isolation is poor, signals from a neighbouring satellite may enter the receiver.
- This is called adjacent satellite interference.
- It is more severe when high-power transponders operate in the same frequency band.

Adjacent Satellite Interference

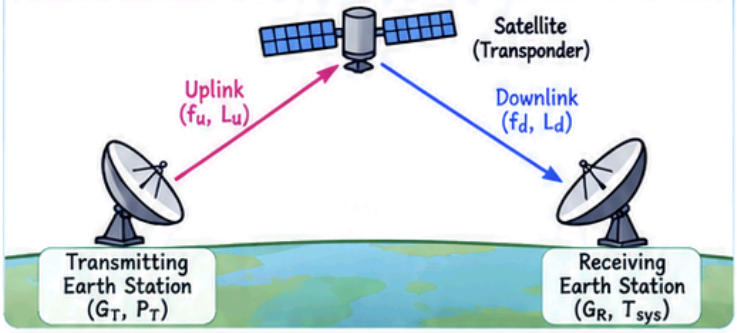


8. Methods to Reduce Interference

1. Proper frequency planning and coordination with ITU.
2. Use of adequate guard bands between channels.
3. High isolation between transmitter and receiver filters.
4. Use of orthogonal polarizations and high XPD.
5. Proper antenna design with high side-lobe suppression.
6. Careful beam and satellite spacing (e.g., 2° in GEO).

Derive the general link equation, obtain the expression for C/n and G/T ratios. Explain the importance of the ratios on satellite link design.

1. Satellite Communication Link



2. Assumptions for Link Equation

- Free space propagation between transmitter, satellite and receiver.
- Line of sight communication.
- Antenna gains are in the direction of each other.
- Losses due to atmosphere, pointing, polarization etc., are included as separate loss terms.
- Satellite transponder is linear with gain G_s and adds its own noise.
- Narrowband communication is considered.

3. Derivation of General Link Equation

The received carrier power at the receiver is given by the Friis transmission formula with all gains and losses included.

$$C = P_T G_T G_s G_R \left(\frac{\lambda}{4\pi R}\right)^2 \frac{1}{L_T L_R L_s L_p L_{atm}}$$

where
- P_T – Transmit power (W)
- G_T – Transmit antenna gain
- G_s – Satellite transponder gain
- G_R – Receive antenna gain
- λ – Wavelength (m)
- R – Distance between terminals (m)
- L_T – Transmit feeder and miscellaneous losses
- L_R – Receive feeder and miscellaneous losses
- L_s – Additional losses (polarization, pointing, etc.)
- L_{atm} – Atmospheric losses

In decibel form,

$$C(dBW) = EIRP + G_R + 20\log\frac{\lambda}{4\pi R} - L_T - L_R - L_s - L_{atm}$$

where

$$EIRP = P_T(dBW) + G_T(dB) + G_s(dB)$$

(Effective Isotropic Radiated Power at satellite output)

4. Derivation of C/n (Carrier to Noise Density Ratio)

Noise power at the receiver is

$$N = kT_{sys}B$$

- k – Boltzmann constant = 1.38×10^{-23} J/K
- T_{sys} – System noise temperature (K)
- B – Receiver noise bandwidth (Hz)

Noise power spectral density (noise density) is

$$n = \frac{N}{B} = kT_{sys} \quad (\text{W/Hz})$$

Therefore, carrier to noise density ratio is

$$\frac{C}{n} = \frac{C}{kT_{sys}} \left(\frac{W}{Hz}\right)$$

In dB form,

$$\frac{C}{n}(dBHz) = C(dBW) - 10\log(kT_{sys})$$

$$= C(dBW) - 10\log(1.38 \times 10^{-23} T_{sys})$$

$$= C(dBW) + 228.6 - 10\log(T_{sys})$$

5. Derivation of G/T Ratio

The figure of merit of a receiving earth station is the ratio of antenna gain to the system noise temperature.

$$\frac{G}{T} = \frac{G_R}{T_{sys}} \left(\frac{1}{K}\right)$$

In dB form,

$$\frac{G}{T}(dB/K) = G_R(dBi) - 10\log(T_{sys})$$

System Noise Temperature (T_{sys})

$$T_{sys} = T_{ant} + T_{rec} + T_{misc}$$

where
- T_{ant} – Antenna noise temperature (due to sky, earth, atmosphere)
- T_{rec} – Receiver noise temperature (LNA, receiver etc.)
- T_{misc} – Additional noise (feeder, losses etc.)

6. Importance of C/n and G/T Ratios in Satellite Link Design

Importance of C/n Ratio

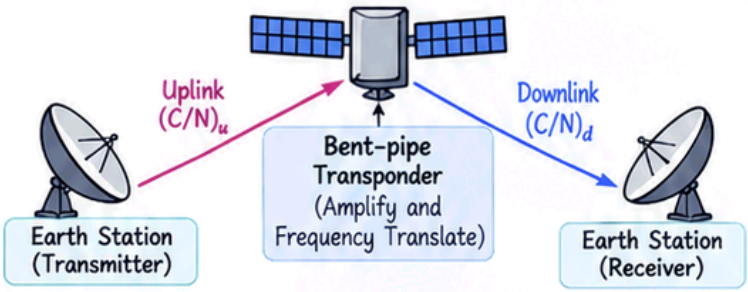
- Indicates the strength of the received carrier relative to the noise level.
- Determines the quality of the received signal.
- Used to evaluate the performance parameters such as:
  - Bit error rate (BER)
  - Carrier to noise ratio (C/N)
  - Carrier to interference ratio (C/I)
- Helps in selecting modulation and coding schemes.
- Ensures reliable communication with minimum error rate.

Importance of G/T Ratio

- Represents the receiving capability of an earth station.
- Higher G/T means better ability to receive weak signals.
- Independent of transmitted power and link distance (used to compare earth stations).
- Helps in link budget analysis and in determining required EIRP from the satellite.
- Used in selecting antenna size, LNA and receiver design.
- Essential for achieving required C/n and C/N for acceptable BER.
- A higher G/T reduces the required satellite transmit power and improves overall link performance.

Derive a formula for the combined C/N ratio of bent-pipe transponder satellite link

1. Bent-Pipe Transponder Satellite Link



2. Noise Sources in a Bent-Pipe Link

- Uplink noise: system noise temperature of transmitting earth station and uplink path.
- Transponder noise: noise added due to the satellite transponder (amplifier).
- Downlink noise: system noise temperature of receiving earth station and downlink path.
- Assume the noises in the three sections are uncorrelated and additive.

3. C/N Ratios of Individual Sections

Uplink C/N ratio $(C/N)_u$

$$\left(\frac{C}{N}\right)_u = \frac{C_u}{N_u} = \frac{EIRP)_u}{kT_{su}B}$$

where

T_{su} – equivalent noise temperature of uplink (transmitting earth station + uplink path)

Transponder C/N ratio $(C/N)_T$

$$\left(\frac{C}{N}\right)_T = \frac{C_T}{N_T} = \frac{P_{in}}{kT_sB}$$

where

T_s – equivalent noise temperature of satellite transponder

P_{in} – input carrier power to the transponder

Downlink C/N ratio $(C/N)_d$

$$\left(\frac{C}{N}\right)_d = \frac{C_d}{N_d} = \frac{EIRP)_s G_R}{kT_{sd}B}$$

where

T_{sd} – equivalent noise temperature of downlink (receiving earth station + downlink path)

4. Derivation of Combined C/N Ratio

- Since the noises in the uplink, transponder and downlink are uncorrelated, the total noise at the receiver is the sum of the individual noise powers referred to the output.
- Let the received carrier power at the output be C.
- Let N_u, N_T and N_d be the noise powers at the output due to uplink, transponder and downlink respectively.
- Then, total noise power $N = N_u + N_T + N_d$
- Therefore, the combined carrier to noise ratio is

$$\left(\frac{C}{N}\right)_{\text{total}} = \frac{C}{N_u + N_T + N_d}$$

Expressing in terms of individual C/N ratios

Since $(C/N)_u = C/N_u \Rightarrow N_u = \dfrac{C}{(C/N)_u}$

$(C/N)_T = C/N_T \Rightarrow N_T = \dfrac{C}{(C/N)_T}$

$(C/N)_d = C/N_d \Rightarrow N_d = \dfrac{C}{(C/N)_d}$

Substituting in total noise,

$$N = C\left[\frac{1}{(C/N)_u} + \frac{1}{(C/N)_T} + \frac{1}{(C/N)_d}\right]$$

5. Combined C/N Ratio Formula

Hence, the combined C/N ratio of a bent-pipe transponder satellite link is

$$\frac{1}{\left(\frac{C}{N}\right)_{\text{total}}} = \frac{1}{\left(\frac{C}{N}\right)_u} + \frac{1}{\left(\frac{C}{N}\right)_T} + \frac{1}{\left(\frac{C}{N}\right)_d}$$

or

$$\left(\frac{C}{N}\right)_{\text{total}} = \frac{1}{\frac{1}{(C/N)_u} + \frac{1}{(C/N)_T} + \frac{1}{(C/N)_d}}$$

6. Notes

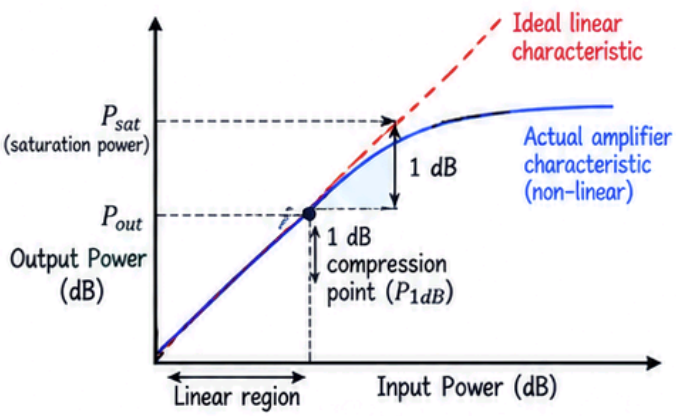
- The above expression is valid for a linear (bent-pipe) transponder where the satellite only amplifies and frequency translates the signal (no demodulation).
- The individual C/N ratios are usually calculated using the link equation for uplink, transponder and downlink.
- The combined C/N ratio shows the overall link quality and is always lower than the smallest individual C/N ratio.
- The link design should ensure that each section provides a sufficiently high C/N so that the overall C/N meets the required value.

Output Back-off used to reduce non-linearity at the satellite output. How is it related to input back-off?

1. Non-linearity in Satellite Transponder

- Satellite transponders use High Power Amplifiers (HPA) (usually Travelling Wave Tube Amplifier – TWTA or Solid State Power Amplifier – SSPA).
- When the input power increases, the output power initially increases linearly.
- After a certain point (saturation), the amplifier becomes non-linear.
- Non-linearity causes:
  - Gain compression
  - Amplitude and phase distortion
  - Spectral regrowth (adjacent channel interference)
- To avoid this, back-off is used.

2. Input-Output Characteristic of HPA



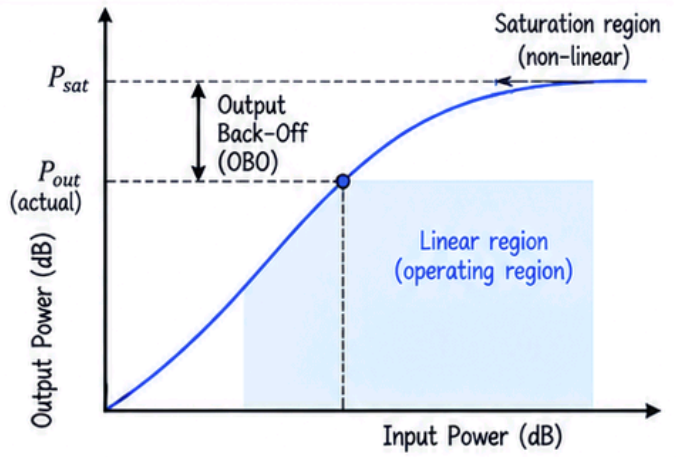
3. Output Back-Off (OBO)

- Output Back-Off (OBO) is the amount by which the actual operating output power is kept below the saturation power of the amplifier.
- It ensures the amplifier operates in the linear region, reducing non-linearity and distortion.
- OBO is usually measured in dB.

$$OBO = P_{sat} - P_{out}\ (dB)$$

where,

P_{sat} – Output saturation power of the HPA (dB)
P_{out} – Actual operating output power (dB)



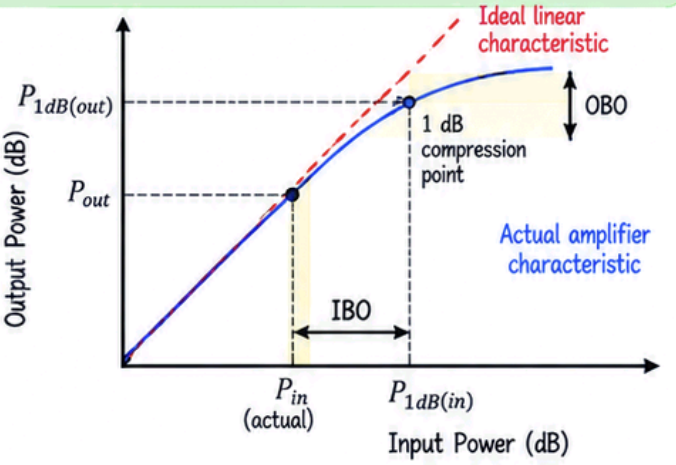
4. Input Back-Off (IBO)

- Input Back-Off (IBO) is the amount by which the actual input power is kept below the input power at the 1 dB compression point.
- It limits the input drive so that the output remains in the linear region.
- IBO is also measured in dB.

$$IBO = P_{1dB(in)} - P_{in}\ (dB)$$

where,

$P_{1dB(in)}$ – Input power at 1 dB compression point (dB)
P_{in} – Actual operating input power (dB)



5. Relation between Output Back-Off and Input Back-Off

- IBO and OBO are related through the small-signal gain (G) of the amplifier in the linear region.
- If G is the small-signal gain (dB), then

$$OBO = IBO + G$$

where,

G – Small-signal power gain of the HPA (dB)

- For a given OBO, the required IBO depends on the amplifier gain.
- Higher gain → smaller IBO for the same OBO.
- This relation is valid in the linear operating region.

6. Importance of Output Back-Off

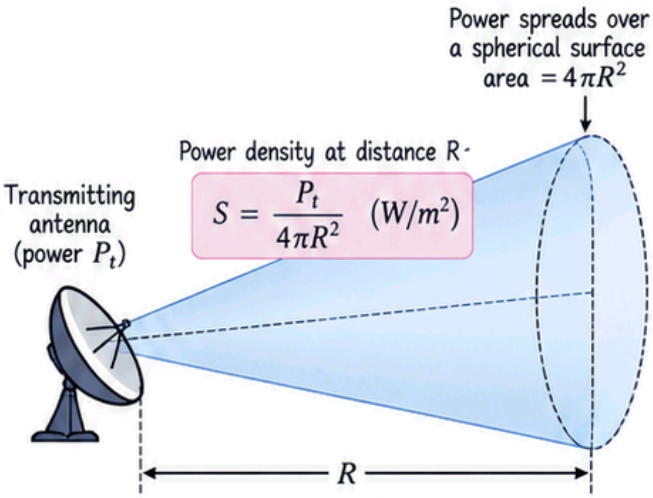
1. Reduces non-linearity and gain compression.
2. Minimizes spectral regrowth and adjacent channel interference.
3. Maintains signal quality (less distortion).
4. Improves linearity for multi-carrier signals.
5. Ensures efficient and reliable operation of the satellite transponder.
6. There is a trade-off: larger OBO gives better linearity but reduces output power and efficiency.

Free Space Transmission Loss and Received Power in dBW

1. Free Space Transmission Loss (FSPL) – Explanation

- Free space transmission loss is the loss in signal power as the electromagnetic wave propagates through space without any physical medium (no obstacles, no reflection, no scattering).
- It occurs due to the spreading of the wavefront over a spherical surface as the distance increases.
- As the signal travels, the same power is distributed over a larger area (surface area of a sphere $= 4\pi R^2$), so the power density decreases with the square of the distance.
- It increases with distance and with frequency.
- It is an important loss in satellite communication links (since most of the path is through free space).

2. Geometry of Free Space Loss



3. Derivation of Free Space Transmission Loss

Power density at a distance R from a transmitting antenna of power P_t is

$$S = \frac{P_t}{4\pi R^2} \quad (W/m^2)$$

The effective aperture area (A_e) of the receiving antenna is

$$A_e = \frac{\lambda^2}{4\pi} \quad \text{(for ideal antenna)}$$

Received power (P_r) is

$$P_r = S \times A_e = \frac{P_t}{4\pi R^2} \times \frac{\lambda^2}{4\pi} = P_t \left(\frac{\lambda}{4\pi R}\right)^2$$

Hence, the free space transmission loss (ratio of transmitted power to received power) is

$$L_{fs} = \frac{P_t}{P_r} = \left(\frac{4\pi R}{\lambda}\right)^2$$

4. Free Space Loss in dB

Free space transmission loss in decibels is

$$L_{fs}(dB) = 10 \log \frac{P_t}{P_r} = 10 \log \left(\frac{4\pi R}{\lambda}\right)^2$$

$$L_{fs}(dB) = 20 \log \left(\frac{4\pi R}{\lambda}\right)$$

Using $\lambda = \frac{c}{f}$,

$$L_{fs}(dB) = 20 \log \left(\frac{4\pi R f}{c}\right)$$

For R in km and f in GHz,

$$L_{fs}(dB) = 92.45 + 20 \log R(km) + 20 \log f(GHz)$$

R	–	distance between transmitting and receiving antennas (m or km)
f	–	frequency of signal (Hz or GHz)
λ	–	wavelength (m)
c	–	speed of light $= 3 \times 10^8$ m/s

5. Received Power in dBW

From the link equation,

$$P_r = P_t \, G_t \, G_r \left(\frac{\lambda}{4\pi R}\right)^2$$

In dB (dBW),

$$P_r(dBW) = P_t(dBW) + G_t(dB) + G_r(dB) - L_{fs}(dB)$$

where

$P_t(dBW)$	–	transmitted power in dBW
G_t	–	transmit antenna gain (dB)
G_r	–	receive antenna gain (dB)
$L_{fs}(dB)$	–	free space transmission loss (dB)

6. Example Calculation

Find the received power for a satellite link with

$$P_t = 10 \text{ dBW}, \quad G_t = 30 \text{ dB}, \quad G_r = 40 \text{ dB}$$
$$R = 36{,}000 \text{ km}, \quad f = 6 \text{ GHz}$$

Step 1: Calculate free space loss

$$\begin{aligned} L_{fs} &= 92.45 + 20 \log R + 20 \log f \\ &= 92.45 + 20 \log (36{,}000) + 20 \log (6) \\ &= 92.45 + 91.13 + 15.56 \end{aligned}$$

$$L_{fs} = 199.14 \text{ dB}$$

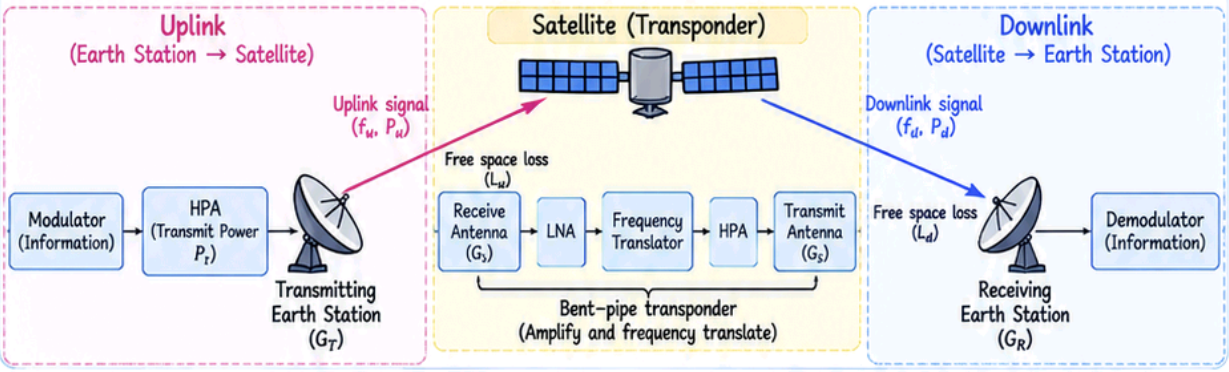
Step 2: Received power

$$P_r(dBW) = 10 + 30 + 40 - 199.14$$

$$P_r = -119.14 \text{ dBW}$$

Power Flow Diagram and Combined Uplink & Downlink Communication

1. Power Flow Diagram of a Satellite Communication Link



2. Power Flow in dB (Uplink and Downlink)

Uplink Power Flow (Transmitter → Satellite)

$$P_t\ (dBW) \rightarrow +G_T \rightarrow -L_u \rightarrow +G_s \rightarrow = C_u\ (dBW)$$

Downlink Power Flow (Satellite → Receiver)

$$P_s\ (dBW) \rightarrow +G_s \rightarrow -L_d \rightarrow +G_R \rightarrow = C_d\ (dBW)$$

3. Combined Uplink and Downlink Link

- Information from the transmitting earth station is sent to the satellite through the uplink.
- The satellite transponder receives the uplink signal, amplifies it and converts it to a different frequency (downlink frequency).
- The amplified signal is then transmitted back to the earth through the downlink.
- The received signal at the destination earth station is the result of both uplink and downlink paths, so both links must be considered together.
- The overall link performance depends on the gains, losses and noise contributions from both uplink and downlink.

4. Link Equations (in dB form)

Uplink Received Carrier Power at Satellite (C_u)

$$C_u\,(dBW) = P_t\,(dBW) + G_T\,(dB) + G_s\,(dB) - L_u\,(dB)$$

where

- P_t – Transmit power at earth station (dBW)
- G_T – Transmit antenna gain (dB)
- G_s – Satellite receive antenna gain (dB)
- L_u – Uplink free space loss (dB)

Downlink Received Carrier Power at Earth Station (C_d)

$$C_d\,(dBW) = P_s\,(dBW) + G_s\,(dB) + G_R\,(dB) - L_d\,(dB)$$

where

- P_s – Satellite transmit power (output of HPA) (dBW)
- G_s – Satellite transmit antenna gain (dB)
- G_R – Receive antenna gain at earth station (dB)
- L_d – Downlink free space loss (dB)

5. Overall (Combined) Link Equation

The overall received carrier power at the destination earth station can be written as

$$C_{total}\,(dBW) = P_t + G_T - L_u + G_s + (Satellite\ gain) + G_R - L_d$$

If satellite gain (transponder) is G_{sat} (dB), then

$$C_{total}\,(dBW) = P_t + G_T + G_{sat} + G_R - (L_u + L_d)$$

where

- P_t – Transmit power at earth station (dBW)
- G_T – Transmit antenna gain (dB)
- G_{sat} – Overall satellite transponder gain (dB) (receive antenna gain + HPA gain – losses)
- G_R – Receive antenna gain at earth station (dB)
- L_u – Uplink free space loss (dB)
- L_d – Downlink free space loss (dB)

6. Explanation of Combined Uplink and Downlink Communication

- **Uplink:** Earth station transmits the signal to the satellite at frequency f_u (e.g., 6 GHz). The signal experiences free space loss and is received by the satellite with reduced power.
- **Transponder:** The satellite amplifies the received signal, converts it to a different frequency (f_d, e.g., 4 GHz) and transmits it back to earth.
- **Downlink:** The amplified signal is radiated towards the destination earth station. After free space loss, it is received and demodulated.
- Both uplink and downlink must be designed to provide sufficient carrier power and C/N ratio for reliable communication.
- The weaker of the two links (uplink or downlink) usually determines the overall link performance.

7. Key Points

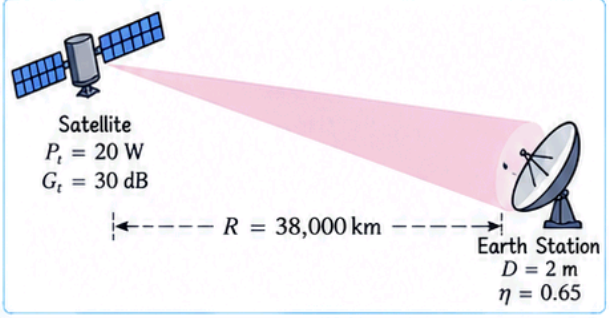
1. The satellite works as a bent-pipe transponder (amplify and frequency translate).
2. Both uplink and downlink have their own free space losses, antenna gains and other losses.
3. The overall received power and C/N ratio depend on the combination of both links.
4. Proper link design ensures that both uplink and downlink provide adequate signal strength and C/N ratio.
5. The link is successful only if both links meet the required performance criteria.

A geostationary satellite carries a C-band transponder which transmits 20 watts into an antenna with an on-axis gain of 30 dB. An earth station is in the center of the antenna beam from the satellite, at 38,000 km. For a frequency of 4.0 GHz:

(i) Calculate the incident flux density at the earth station in watts per square meter and in dBW/m^2.
(ii) The earth station has an antenna with a circular aperture 2 m in diameter and an aperture efficiency of 65%. Calculate the received power level in watts and in dBW at the antenna output port.
(iii) Calculate the on-axis gain of the antenna in dB.
(iv) Calculate the free space path loss between the satellite and the earth station.
(v) Calculate the power received, P_r, at the earth station.

Given Data

Transmit power (satellite)	P_t	$= 20$ W
Satellite antenna gain	G_t	$= 30$ dB
Distance (geostationary)	R	$= 38{,}000\,\text{km} = 3.8\times10^7$ m
Frequency	f	$= 4.0\,\text{GHz} = 4\times10^9$ Hz
Earth station antenna diameter	D	$= 2$ m
Aperture efficiency	η	$= 65\% = 0.65$



(i) Incident flux density at the earth station

Flux density at distance R is

$$S = \frac{P_t G_t}{4\pi R^2} \quad (W/m^2)$$

$P_t = 20$ W, $\quad G_t = 10^{30/10} = 1000$

$R = 3.8\times10^7$ m

$$S = \frac{20\times1000}{4\pi\,(3.8\times10^7)^2}$$

$$= \frac{20000}{1.8136\times10^{16}}$$

$$S = 1.10\times10^{-12}\ W/m^2$$

In dBW/m^2

$$S\,(dBW/m^2) = 10\log_{10}(S)$$

$$= 10\log_{10}(1.10\times10^{-12})$$

$$= 10\,(-11.96)$$

$$S = -119.6\ \text{dBW/m}^2$$

(ii) Received power at the earth station antenna output port

Effective aperture area of the earth station antenna

$$A_e = \eta\frac{\pi D^2}{4}$$

$D = 2$ m, $\quad \eta = 0.65$

$$A_e = 0.65\times\frac{\pi\,(2)^2}{4} = 0.65\times\pi = 2.042\ \text{m}^2$$

Received power

$$P_r = S\times A_e$$

$$P_r = 1.10\times10^{-12}\times2.042$$

$$P_r = 2.25\times10^{-12}\ W$$

In dBW

$$P_r\,(dBW) = 10\log_{10}(P_r)$$

$$= 10\log_{10}(2.25\times10^{-12})$$

$$= 10\,(-11.65)$$

$$P_r = -116.5\ \text{dBW}$$

(iii) On-axis gain of the earth station antenna

Antenna gain is

$$G_r = \eta\left(\frac{\pi D}{\lambda}\right)^2$$

Wavelength at 4.0 GHz:

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{4\times10^9} = 0.075\ \text{m}$$

$$G_r = 0.65\left(\frac{\pi\times2}{0.075}\right)^2$$

$$= 0.65\,(83.78)^2$$

$$= 0.65\times7029.7$$

$$= 4569$$

In dB

$$G_r\,(dB) = 10\log_{10}(4569)$$

$$= 10\times3.66$$

$$G_r = 36.6\ \text{dB}$$

(iv) Free space path loss between satellite and earth station

Free space loss is

$$L_{fs}(dB) = 92.45 + 20\log_{10}(R_{km}) + 20\log_{10}(f_{GHz})$$

$R = 38{,}000$ km, $\quad f = 4.0$ GHz

$$L_{fs} = 92.45 + 20\log_{10}(38{,}000) + 20\log_{10}(4.0)$$

$$= 92.45 + 91.60 + 12.04$$

$$L_{fs} = 196.1\ \text{dB}$$

(Alternative formula)

$$L_{fs} = \left(\frac{4\pi R}{\lambda}\right)^2$$

$$L_{fs}\,(dB) = 20\log_{10}\left(\frac{4\pi R}{\lambda}\right)$$

$= 196.1$ dB (same result)

(v) Power received at the earth station using link equation

Link equation:

$$P_r(dBW) = P_t(dBW) + G_t(dB) + G_r(dB) - L_{fs}(dB)$$

$$P_r(dBW) = 10\log_{10}(20) = 13.01\ dBW$$

$$P_r(dBW) = 13.01 + 30 + 36.6 - 196.1 = -116.5\ dBW$$

$$P_r = 10^{-116.5/10}$$

$$= 2.25\times10^{-12}\ \text{W}$$

(same as part ii)